Pediatric Patient Counting with Interactive Object Detection

Peter Amoah Mensah (Index: 3457122)
Fafali Dorkunor (Index: 3455022)

Department of Mathematics
Kwame Nkrumah University of Science and Technology

February 20, 2026

Abstract

The accurate enumeration of pediatric patients in resource-constrained hospital environments remains a significant challenge due to the frequent occlusion of infants carried by caregivers. This project bridges the gap between administrative hospital records and physical reality through the application of Relational Artificial Intelligence and mathematical optimization. We propose an automated, privacy-preserving Edge computing architecture that utilizes a multi-stage algorithmic approach grounded in mathematical anthropometry, topological graph matching, and geometric centroid tracking. By calculating dimensionless Head-to-Body Ratios and evaluating spatial invariants, the system successfully bypasses traditional non-maximum suppression limits to detect occluded children. Operating exclusively in volatile memory, the framework ensures strict patient confidentiality while outputting actionable demographic data to optimize the clinical allocation of pediatric resources.

1 Introduction

This project is concerned with tracking the accurate number of young children visiting a hospital, particularly those who might not be directly registered as primary patients. In major Ghanaian hospitals, such as Komfo Anokye Teaching Hospital (KATH), optimizing medical staff allocation requires precise demographic data. Cultural practices dictate that women carry babies or toddlers strapped to their backs while navigating crowded Outpatient Departments (OPDs). However, electronic health records (e.g., LHIMS) systematically capture only the primary patient, leaving accompanying family members statistically invisible (Addotey-Delove et al., 2023).

The clinical implications of this "invisible child" phenomenon are severe. As highlighted in modern pediatric treatment protocols, children are not merely "small adults." The pediatric demographic requires highly specialized clinical care, including age-specific developmental pharmacokinetics, precise allometric scaling for weight-banded dosing, and targeted formulations. This undocumented influx skews hospital statistics, exacerbating OPD wait times (which can average 2–4 hours) and causing dangerous delays in triage (Osei et al., 2024). Our goal is to design a mathematical computer vision system that can detect, differentiate, and count children entering the hospital, ensuring that every infant is acknowledged as part of the facility's active demographic footprint.

2 Problem Statement

Currently, hospitals rely heavily on manual tallying or flawed official records to estimate their pediatric load. This method is highly susceptible to human error and lacks the real-time statistical reliability needed for dynamic hospital operations.

While the natural mathematical solution is automated Computer Vision, introducing standard surveillance algorithms to a Ghanaian OPD creates a severe geometric flaw (Ouyang & Wang, 2015). Because infants are carried on the back, their spatial coordinates heavily intersect with the mother's. Spatial bounding algorithms interpret the overlapping geometries as a single adult entity. Standard Non-Maximum Suppression (NMS) actively merges these overlapping shapes, discarding the carried child's features as visual noise. Consequently, administrators cannot properly allocate pediatric triage spaces or ensure sufficient clinical supplies. Furthermore, standard surveillance attempts often store video data on centralized servers, violating the strict patient privacy standards required in medical environments.

3 Objectives

The primary objective of this study is to develop a privacy-preserving, automated mathematical framework capable of accurately counting pediatric visitors, including carried infants. Specifically, the project aims to:

- **Automated Child Detection:** Formulate a spatial detection model to mathematically separate carried children from adult caregivers, successfully bypassing occlusion failures.

- **Mathematical Anthropometry:** Apply physical constraints via Head-to-Body Ratios to prevent false positives, such as misclassifying short adults.
- **Data Tracking and Re-Identification:** Implement geometric centroid tracking utilizing Euclidean distance to ensure unique, non-repeating patient counts across temporal frames.
- **Privacy-Preserving Edge Architecture:** Design a localized computing architecture that outputs strictly numerical matrices without storing visual data, guaranteeing absolute patient privacy.

4 Methodology

We explore a mathematical, technology-driven approach utilizing advanced computer vision and spatial inference. Continuous video feeds from standard OPD surveillance cameras will be processed locally on Edge hardware.

Camera-based Child Differentiation (Allometric Scaling): To accurately classify a person's age regardless of their distance from the camera lens, the system utilizes mathematical anthropometry (Mlinarić et al., 2024). Specifically, it calculates a scale-invariant Head-to-Body Ratio (R):

$$R = \frac{H_{\text{total}}}{H_{\text{head}}}$$

where H_{total} is the subject's total bounding height and H_{head} is their isolated head height. Based on human growth allometry, adults yield $R \approx 7.5 - 8.0$, whereas infants and young children yield $R \approx 4.0 - 6.0$. This dimensionless ratio acts as a mathematical constant, physically preventing short adults from being statistically misclassified as children.

Detecting Carried Children (Spatial Inference): To solve the occlusion problem, the proposed algorithm independently isolates all human Heads (H) and Bodies (B). When a single body geometry bounds two distinct heads ($h_1, h_2 \in H$), the system evaluates two spatial parameters:

- **Vertical Displacement:** $y_{h2} < y_{h1}$ (The secondary head is physically lower).
- **Area Proportion:** $A_{h2} < A_{h1}$ (The secondary head's geometric area is strictly smaller).

If both conditions hold true, the algorithm overrides standard suppression limits, correctly infers a carried child, and updates the pediatric matrix by $+1$.

Geometric Tracking: To prevent double-counting when a child leaves the camera's view and quickly returns, their spatial kinematics are tracked. In every video frame t, the system extracts the bounding box centroid $C_t = (x_t, y_t)$. The algorithm links the child across frames by minimizing the Euclidean distance d:

$$d = \sqrt{(x_{t+1} - x_t)^2 + (y_{t+1} - y_t)^2}$$

Through memory reassignment, if a child leaves the frame, the system temporarily caches their exact boundary coordinates to restore their ID and prevent a false count upon re-entry.

Privacy Protocol: To guarantee patient confidentiality, mathematical analysis occurs entirely in volatile memory (RAM). No video or images are ever saved. The system only transmits an anonymized numerical JSON array, such as:


```
{ "timestamp": "10:14:02", "adults": 12, "children": 5, "occluded_infants":
2 }
```


5 Justification

Counting pediatric visitors is far more than an administrative exercise; it is a critical necessity for optimized healthcare delivery. Clinical pediatric care is fundamentally distinct from adult care. By implementing this system, a hospital transitions from estimating its pediatric load to actively quantifying it. Technologically, this project is justified because it adapts complex spatial algorithms to correct occlusion failures in low-resource settings. Culturally, bringing children to outpatient departments is a deeply ingrained reality in Ghana; capturing this demographic mathematically is the necessary first step toward designing genuinely child-responsive healthcare facilities without violating medical privacy.

6 Expected Results

By the conclusion of this project, we expect to deploy a mathematical computer vision system that reliably transforms "invisible" hospital visitors into actionable, quantitative data. We aim to achieve high-precision classification, successfully capturing over 90% of actual children, including those actively carried by caregivers.

The primary deliverable is an automated, real-time dashboard reflecting demographic occupancy loads. We expect this system to integrate directly into hospital operations: for example, if the mathematical pediatric density exceeds a set capacity threshold (e.g., $> 30\%$ occupancy), an operational trigger will alert administrators to open dedicated pediatric triage rooms. The system's precision will be calibrated against physical manual counts and mathematically evaluated using the Mean Absolute Percentage Error (MAPE):

$$MAPE = \frac{1}{n} \sum_{i=1}^{n} \left| \frac{\text{Actual}_i - \text{Predicted}_i}{\text{Actual}_i} \right| \times 100$$

Ultimately, the success of this project will be measured by its ability to provide healthcare administrators with the granular demographic intelligence required to optimize the clinical environment.

References

[1] Addotey-Delove, M. N., et al. (2023). *Adoption of health information systems in emerging economies: Evidence from Ghana*. Global Health Informatics.

[2] Cressey, T. (2021). *Clinical Aspects of Pediatric Treatment*. Academic Medical Education.

[3] Koshkina, M., et al. (2025). *Towards long-term player tracking with graph hierarchies and domain-specific features*. Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV).

[4] Mlinarić, T. J., et al. (2024). *Research on Children's Body Proportions: Determining the Canon of Head Length to Total Body Height*. Applied Sciences, 14(16), 7185.

[5] Osei, E., et al. (2024). *Patient satisfaction with quality of care at out-patient departments in selected health facilities in Kumasi, Ghana*. SpringerMedizin.

[6] Ouyang, W., & Wang, X. (2015). *Partial Occlusion Handling in Pedestrian Detection with a Deep Model*. IEEE Transactions on Circuits and Systems for Video Technology.

[7] Ralhan, P., et al. (2023). *Object detection with centroid-based tracking*. Hackster.io.